

Nathan Tsao

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Applied ML engineer with a background in mechanical engineering and hands-on experience in generative models, reinforcement learning, and scientific machine learning. Passionate about building AI tools that automate real-world decision-making and improve user experiences; project code available at nathantsao.com.

EDUCATION

University of Texas at Austin <i>Masters of Science:</i> Mechanical Engineering <i>Thesis:</i> Neural Port-Hamiltonian Differential Algebraic Equations	Aug 2023 – May 2025 GPA: 3.74
University of Illinois Urbana-Champaign <i>Bachelors of Science:</i> Mechanical Engineering	Aug 2019 – May 2022 GPA: 3.87

WORK EXPERIENCE

NASA Ames Research Center Autonomous Aircraft Operations Research Intern - Developing a decentralized path-planning framework to promote traffic-following behavior among multi-agent autonomous aircraft, enhancing overall safety and operational efficiency. - Using reinforcement learning with RLlib to encourage adaptive traffic following and minimize travel time.	June 2025 – Aug 2025 Mountain View, CA
Berkeley Lights Hardware Engineering Intern - Automated temperature calibration and data-acquisition pipelines in Python. - Developed mechanical and electrical design of temperature calibration sensor of OptoSelect chip.	June 2022 – Sep 2022 Berkeley, CA

RESEARCH EXPERIENCE

Autonomous Systems Group: UT Austin Graduate Research Assistant - Innovated a compositional machine learning modeling framework for differential-algebraic systems, enabling scalable modeling of electrical networks. - Devised a low-power, low-latency human activity recognition algorithm optimized to deploy on batteryless energy-harvesting sensor platforms, resulting in 15-50% relative improvement over baselines.	Dec 2023 – May 2025 Austin, TX
Hybrid Robotics Group: UC Berkeley Visiting Research Assistant - Implemented a proof-of-concept reinforcement learning-based locomotion balancing controller for tailed quadruped robots. - Designed and built hardware for a 3D-printed tail that attaches to a quadruped robot, including custom quasi-direct-drive actuators and FOC motor controller.	May 2022 – Jan 2023 Berkeley, CA
RoboDesign Lab: UIUC Undergraduate Research Assistant - Prototyped a low-cost force-sensing humanoid robot foot using elastomers and Hall sensors. - Explored the use of Gaussian processes to estimate force from elastomer deformation.	Jan 2022 – May 2022 Urbana, IL

PUBLICATIONS

Cyrus Neary*, [Nathan Tsao](#)*, and Ufuk Topcu. Neural Port-Hamiltonian Differential Algebraic Equations for Compositional Learning of Electrical Networks. *Accepted to CDC 2025*.

Geffen Cooper*, [Nathan Tsao](#)*, Filippos Fotiadis, Ufuk Topcu, Radu Marculescu. Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors. *Under review at NeurIPS 2025*.

SELECTED PROJECTS

Causal Diffusion Guidance

May 2025

Statistical Machine Learning Final Project

- Developed a diffusion-based framework for generating counterfactual samples from structural causal models.
- Preliminary experiments on causal image datasets demonstrate the ability to generate high-quality, causally consistent counterfactual samples, suggesting the emergence of causal reasoning in diffusion models.

TEACHING EXPERIENCE

ASE 370C Feedback Control Systems: UT Austin

Jan 2025 – May 2025

Graduate Teaching Assistant

Austin, TX

ME 314D Dynamics: UT Austin

Sep 2023 – Dec 2023

Graduate Teaching Assistant

Austin, TX

HONORS AND AWARDS

Dr. J. Parker Lamb Endowed Presidential Fellowship: UT Austin

2023

Nominated by the Walker Department of Mechanical Engineering.

Highest Honors: UIUC

2022

Awarded to students with at least a 3.8 Illinois GPA and continual commitment to service and education.

Dean's List: UIUC

2019-22

Awarded to undergraduate students in the top 20 percent of their college class.

OUTREACH

Del Valle High School Visit

Mar 2025

Presented machine learning academic research to engineering and robotics high school students.

Austin, TX

UT Austin Girls Day

Feb 2024

Guided K-8th graders to program remote-controlled cars in Python.

Austin, TX

SKILLS

Skills: Machine Learning | Generative AI | Large Language Models | Scientific ML | Linux

Programming Languages: Python | SQL | Dart

Tooling: Git | Docker | Supabase

Frameworks: PyTorch | JAX | Flax | Hugging Face | FastAPI | Pandas | Ray (RLlib) | Stable-Baselines3

Languages: English | Mandarin

Relevant Coursework: Reinforcement Learning | Statistical Machine Learning | Theoretical Statistics | Convex Optimization